RIANSARES 4R
Reformas - Rehabilitatión
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TECHNICAL DOCUMENTATION

APPLIED AI · RAG

2026 EDITION

Insulation and energy renovation — informed decisions

REFERENCE DOCUMENTATION FOR THE AI ASSISTANT

RIANSARES 4R

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riansares.com **athagon.tech**

Insulation and energy renovation — informed decisions

Which should be addressed first: windows or façade?

It depends on actual losses, thermal bridges, ventilation and building condition. An energy assessment helps prioritise and avoids interventions that cause condensation or perform below expectations.

Can internal insulation be installed without losing ventilation?

Yes, with appropriate design, but it changes hygrothermal behaviour and reduces usable floor area. Vapour barriers must be assessed according to the solution and condensation risk, while maintaining adequate ventilation.

Can very airtight windows increase mould?

If ventilation is insufficient, accumulated indoor moisture may increase. The solution is to check air renewal and thermal bridges, rather than avoid efficient windows.

What should be checked before insulating a roof?

Watertightness and substrate condition, loads, thermal continuity, condensation, access and compatibility with the existing system. Insulation does not replace repairing leaks.

Is there a solution that guarantees 50 % savings?

No percentage can be guaranteed without knowing baseline consumption, climate, use, floor area, services and calculation results. Measures and expected savings must be estimated individually.

What is an energy audit for?

It characterises the home and guides improvements to the envelope, equipment and use. A commercial list of materials must not be confused with a documented energy assessment.

Can an energy rating change after renovation?

It may improve if interventions reduce demand and/or consumption under the applicable certification methodology. The specific improvement requires professional calculation and verification.

What are thermal bridges?

Areas where heat flow differs from the rest of the envelope, such as structural junctions or opening perimeters. They can affect losses and condensation; solutions must be compatible with the building.

Is installing insulation on a damp wall sufficient?

No. First identify water entry or condensation and ensure the proposed system is compatible. Covering unresolved damp can cause hidden damage.

How can priorities be chosen on a limited budget?

Compare urgent safety and health needs, envelope condition, energy losses and a reasoned cost/benefit assessment. Do not recommend expensive solutions based solely on trends or photographs.

Case E-11 — Home with cold walls

Hypothetical assessment case, not completed work or a service committed to by Riansares 4R.

Situation: Indoor condensation and recently replaced windows.

Checks: Measure humidity, ventilation and thermal bridges.

Possible scope: Comprehensive envelope and ventilation treatment.

Decision criterion: Do not automatically blame the new glazing.

Evidence to retain: Diagnosis and subsequent verification.

Case E-12 — Very hot space beneath a roof

Hypothetical assessment case, not completed work or a service committed to by Riansares 4R.

Situation: The top floor overheats in summer.

Checks: Existing insulation, shading, ventilation and roof.

Possible scope: Options compared using thermal criteria.

Decision criterion: Do not promise a final temperature without simulation.

Evidence to retain: Technical report and solution data sheets.

Industry references and scope

- CTE_HE.
- IDAE.
- CTE_HS.

The references provide general context and do not establish that Riansares 4R has carried out work. Technical requirements must be verified for the applicable type of intervention and jurisdiction.